

Human Reproduction

Year 9 Science · exam sprint · 2 hours

0:04 Loop 1 · Sex cells and the two systems

0:26 Loop 2 · Hormones and the menstrual cycle

0:49 Loop 3 · Fertilisation, chromosomes, twins

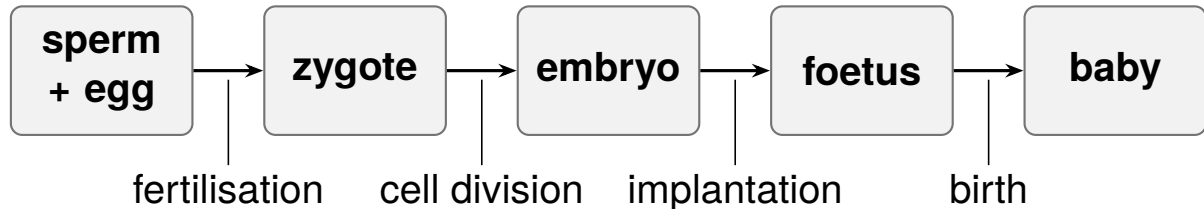
1:12 Loop 4 · Pregnancy, birth, fertility

1:44 Mini mock

Do Now

- 1 Name the male sex cell and the female sex cell.
- 2 Which organ makes the female sex cells?
- 3 What is the joining of the two sex cells called?
- 4 How many chromosomes are in a human body cell?
- 5 Which organ passes oxygen from the mother to the foetus?

The one idea: from two cells to a baby

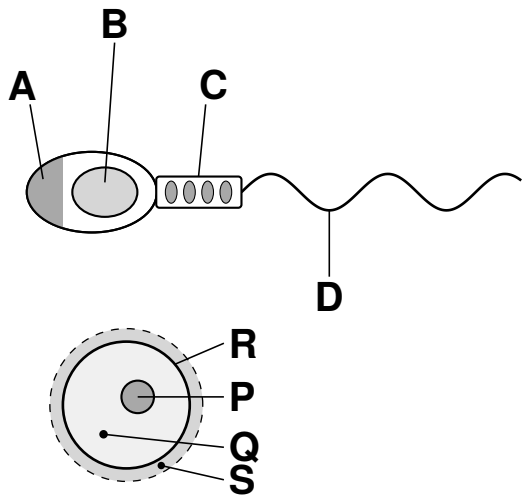


Every question is about one of these steps.

For each step, know the cell, the place and the process.

Which step happens in the oviduct?

Quick review: Sex cells



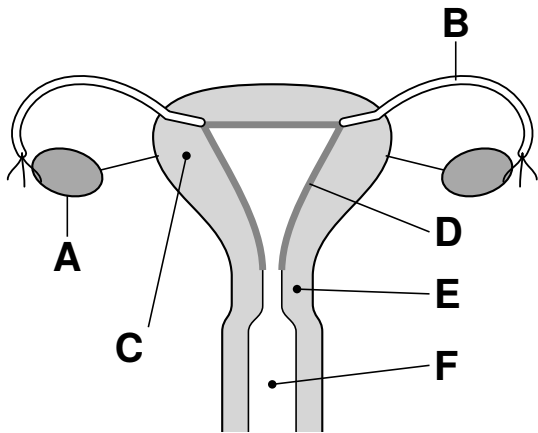
- 1 Name parts A to D and parts P to S.
- 2 How many chromosomes are in each nucleus?
- 3 Which cell is larger?
Which cell can move?

Adaptations: the feature and how it helps

Cell	Feature	How it helps
sperm	long tail	_____
sperm	many mitochondria	_____
sperm	enzymes in the head	_____
egg	large cytoplasm	_____
egg	membrane changes	_____

Why does a man make millions of sperm, but a woman releases one egg a month?

Quick review: Female system, front view

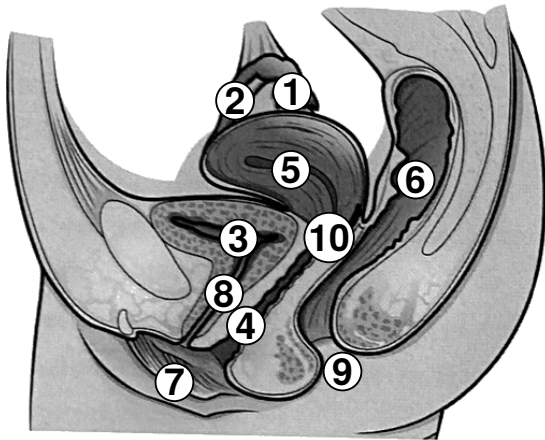


- 1 Name parts A to F.
- 2 Where are eggs made?
- 3 Where does a foetus develop?
- 4 Which part is a ring of muscle?

Female system: the job of each part

Part	Job
ovary	makes _____ and two hormones
oviduct	carries the egg; _____ happens here
uterus	the _____ develops here
lining	gets thick so that an _____ can implant
cervix	a ring of _____ at the base of the uterus
vagina	receives _____; the baby is born through it

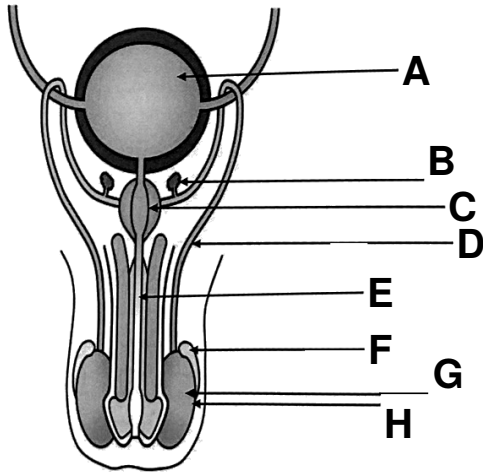
Quick review: Female system, side view



anus · bladder · cervix ·
fallopian tube · ovary · rectum ·
urethra · uterus · vagina · vulva

- 1 Match the words to parts 1 to 10.
- 2 Which four parts are not reproductive organs?

Quick review: Male system



- 1 Name parts A to H.
- 2 Where are sperm made?
- 3 Which tube carries both urine and semen?

The path of a sperm cell

Put the cards in order.

- A** The urethra carries semen out through the penis.
- B** Sperm are made in a testis.
- C** Sperm travel along the sperm duct.
- D** Sperm are stored in the epididymis.
- E** Glands add fluid: sperm + fluid = _____.

Why are the testes held outside the body?

Male system: the job of each part

Part	Job
scrotum	holds the testes outside the body, so they stay _____
testes	make _____ and the hormone _____
epididymis	_____ the sperm
sperm duct	carries sperm to the _____
glands	add _____ that feeds the sperm
penis	puts semen into the _____

Spot the error (1)

Describe where sex cells are made and where a foetus develops.

Sam's answer

Eggs are made in the uterus and travel down the oviduct. The cervix is the place where the foetus develops. In a man, sperm are made in the testes and stored in the scrotum.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

Q1 State two differences between a sperm cell and an egg cell. **[A]**

Q2 Explain why the testes are held outside the body in the scrotum. **[M]**

Q3 A woman's oviducts are both blocked. She still releases an egg each month. Explain why she cannot become pregnant naturally. **[E]**

Classwork

In-class work · Part A

- 1 Label the female reproductive system.
- 2 Complete the sentences about the male system.
- 3 Explain one adaptation of a sperm cell.

Write in full sentences. Use the science words.

Puberty: hormones switch it on

Hormones are chemical _____, carried in the _____.

Boys

Girls

hormone _____

made in testes

ovaries

changes voice deepens

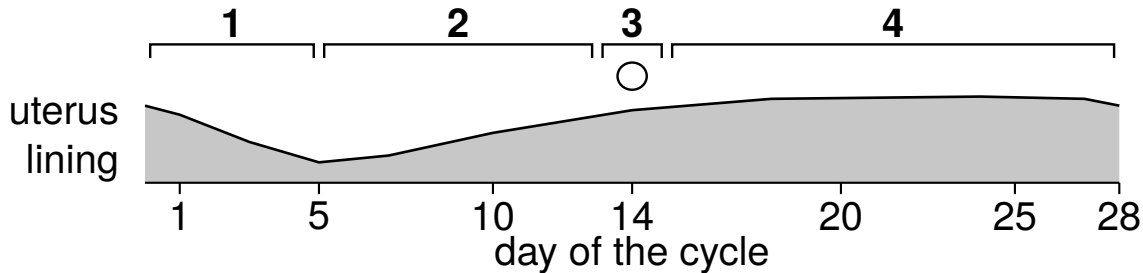
_____ start

also sperm are made

breasts grow

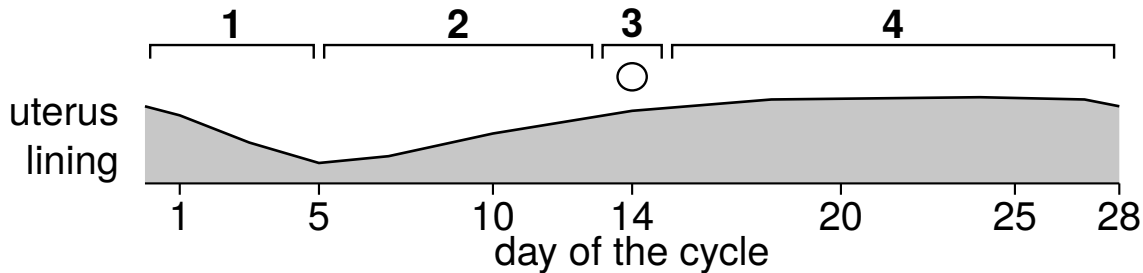
True or false? Girls usually start puberty before boys.

The menstrual cycle: stages 1 and 2



- 1 Days 1 to 5: the lining _____ and leaves through the vagina. This is a _____.
- 2 Days 5 to 13: the lining _____ again. An egg matures in an _____.

The menstrual cycle: stages 3 and 4



3 About day 14: an egg is _____ from an ovary. This is _____.

4 Days 15 to 28: the lining stays _____, ready for a fertilised egg.

When is a woman most likely to become pregnant?

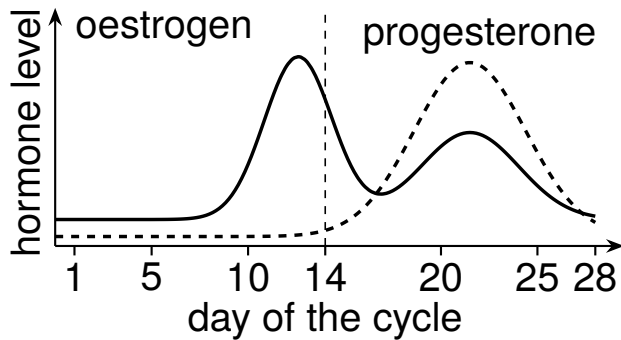
Four hormones, four jobs

Hormone	Made in	Job
FSH	pituitary gland	makes an egg _____ in the ovary
LH	_____	triggers _____ on day 14
oestrogen	ovary	_____ the lining of the uterus
progesterone	_____	keeps the lining _____

FSH = follicle stimulating hormone. LH = luteinising hormone.

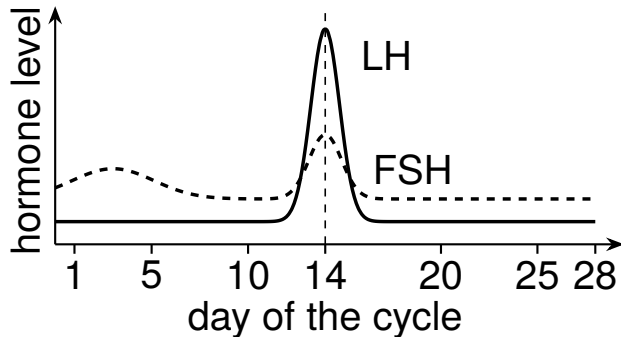
Which two hormones peak at ovulation?

Read the graph: ovarian hormones



- 1 Which hormone peaks just before day 14?
- 2 Which hormone is highest at about day 21?
- 3 Both levels fall by day 28. What happens to the lining?

Read the graph: pituitary hormones



- 1 On which day do LH and FSH peak?
- 2 What does the LH peak cause?
- 3 Describe the level of LH from day 1 to day 12.

No fertilised egg: write it in this order

- 1 The egg is not fertilised, so no embryo _____ in the lining.
- 2 The level of _____ falls.
- 3 The lining of the uterus _____.
- 4 Blood and lining leave through the _____: a period.

A: lines 3–4 **M:** links the hormone to the lining **E:** the whole chain

What is different if the egg is fertilised?

Spot the error (2)

Describe ovulation and name one hormone that changes the lining of the uterus.

Sam's answer

Ovulation is when the lining of the uterus is lost. An egg is released on about day 28 of the cycle. Oestrogen is made in the pituitary gland and makes the lining thicker.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

- Q1** A woman has a 28-day cycle. On which days is she most likely to become pregnant? **[A]**
- Q2** Describe how the lining of the uterus changes between day 5 and day 14. Name the hormone that causes this. **[M]**
- Q3** The level of progesterone stays high in a pregnant woman. Explain why this is important. **[E]**

Classwork

In-class work · Part B

- 1 Name the four stages of the cycle.
- 2 Match each hormone to its job.
- 3 Read a table of hormone levels and explain it.

For data: describe the trend first, with numbers.

From intercourse to fertilisation

Put the cards in order.

- A** Sperm swim through the cervix and uterus into the oviducts.
- B** The penis fills with blood and becomes erect.
- C** One sperm enters the egg, and the two nuclei fuse.
- D** Semen is released into the vagina. This is ejaculation.
- E** The penis is put into the vagina.

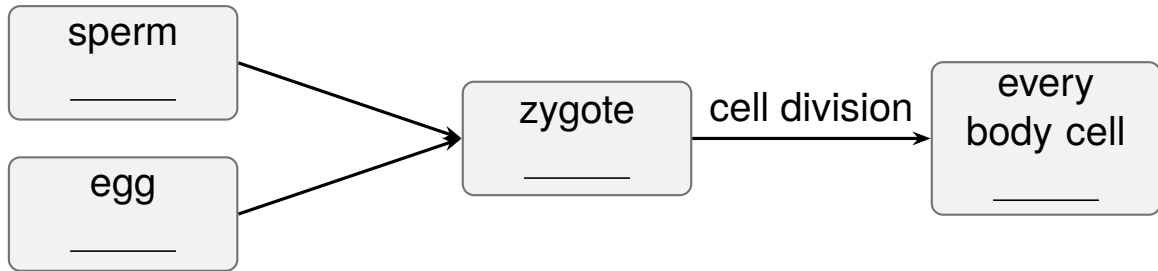
Which card is fertilisation? Where does it happen?

Zygote, embryo, foetus

- 1 The sperm nucleus fuses with the egg nucleus. The new cell is a _____.
- 2 It divides many times as it moves along the oviduct. The ball of cells is an _____.
- 3 The ball of cells sinks into the lining of the uterus. This is _____.
- 4 After about 8 weeks it has all its organs. Now it is a _____.

What stops a second sperm from entering the egg?

Chromosomes: count them



Chromosomes carry the genes. They are in the _____ of every cell.

Why must a sex cell have only half the number?

Boy or girl?

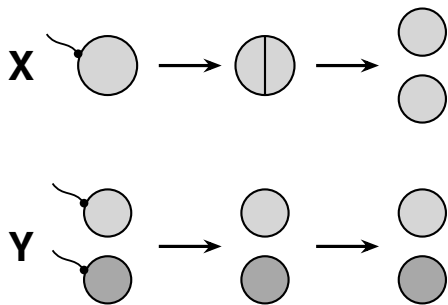
		mother's eggs	
		X	X
father's sperm	X	_____	_____
	Y	_____	_____

Female cells have XX. Male cells have XY.

Every egg carries _____. A sperm carries _____ or _____.

- 1 Complete the square.
- 2 What is the chance of a boy?
- 3 Which parent's sex cell decides?

Twins



Each row ends with two embryos.

1 Which row shows identical twins?

2 Row X: _____ egg, _____ sperm.

The zygote _____.

3 Row Y: _____ eggs, _____ sperm.

4 Which twins can be a boy and a girl?

Workbook: true or false?

- a** Identical twins happen when a fertilised egg cell divides in two, but the two cells split apart.
- b** Identical twins will always be different sexes.
- c** Non-identical twins happen when two egg cells are fertilised.
- d** Non-identical twins can be the same sex or different sexes.
- e** Identical twins happen when two egg cells are fertilised.
- f** Non-identical twins happen when egg cells turn into embryos on their own.

Workbook: fertilisation

Q1 Where does fertilisation occur?

Q2 Only one sperm cell joins with the egg cell. What do you think happens to all the other sperms?

Q3 Humans have internal fertilisation as a result of moving onto the land in our evolutionary history. Why do you think this is an advantage on the land?

A boy next time? Write it in this order

A couple have three girls. What is the chance that their fourth child is a boy?

- 1 Every egg carries an _____ chromosome.
- 2 Half of the sperm carry X and half carry _____.
- 3 So at every fertilisation the chance of a boy is _____.
- 4 The children born before do not change the _____.

A: the number **M:** lines 1–2 **E:** line 4 as well

Spot the error (3)

Explain where a baby's chromosomes come from, and how twins can be identical.

Sam's answer

A sperm has 46 chromosomes and an egg has 46, so the baby gets all of them. The mother's egg decides if the baby is a boy or a girl. Identical twins form when two eggs are fertilised by two sperm.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

- Q1** State where fertilisation happens and name the cell that is formed. **[A]**
- Q2** A boy and a girl are twins. Are they identical twins? Explain. **[M]**
- Q3** Explain why a child has features of both parents but is not identical to either of them. **[E]**

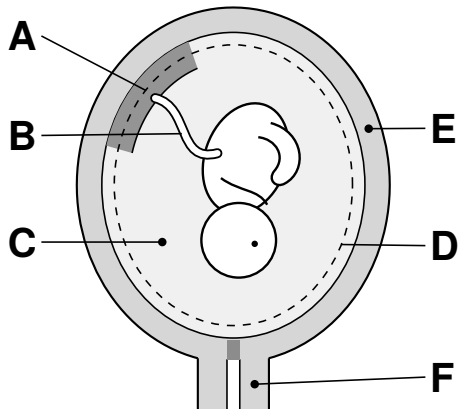
Classwork

In-class work · Part C

- 1 Put the stages from sex cell to foetus in order.
- 2 Work out the sex of a baby from its chromosomes.
- 3 Explain how two kinds of twins form.

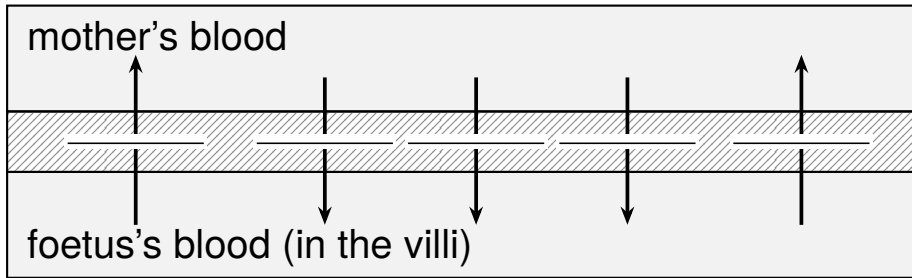
Give the number of chromosomes each time you name a cell.

Quick review: The foetus in the uterus



- 1 Name parts A to F.
- 2 Which part cushions the foetus?
- 3 Which part joins the foetus to the placenta?

The placenta: what crosses?



Down: three things the foetus needs. Up: two waste products.

The two bloods never _____. Substances cross by _____.

Workbook: the role of the placenta

The placenta develops from foetal tissues and allows the _____ of materials between mother and foetus. It has a large number of blood vessels that do not touch. This ensures there is no _____ of the maternal and foetal blood.

- a** Give three reasons why it is important that the maternal and foetal blood do not mix.
- b** Explain why the embryo needs oxygen and glucose.
- c** Which substances diffuse from the foetus' blood into the mother's blood and why is it important that they are removed?

Oxygen to the foetus: write it in this order

- 1 The mother's blood has a _____ concentration of oxygen than the foetus's blood.
- 2 Oxygen _____ from high to low concentration, across the thin wall.
- 3 The villi give a large _____, so diffusion is fast.
- 4 The foetus's blood carries the oxygen away in the umbilical _____.

A: line 2 **M:** lines 1–2 **E:** lines 3–4 as well

Now tell the same story for carbon dioxide.

Harm: what else crosses the placenta?

From the mother	Effect on the foetus
-----------------	----------------------

cigarette smoke	less _____ reaches it; the baby is smaller
alcohol	damages the developing _____
drugs	the baby can be born addicted
rubella virus	deafness; heart and eye defects

Why can even a small amount do a lot of damage?

Pregnancy in numbers

- 1 Pregnancy (gestation) lasts about _____ weeks, counted from the first day of the last period.
- 2 The embryo is called a foetus from about week _____.
- 3 Losing a foetus before _____ weeks is a miscarriage.
- 4 A baby born before _____ weeks is premature.

A baby is born at 35 weeks. Which word describes this birth?

Birth: three stages

Put the cards in order. Then name each stage.

A The placenta is pushed out.

B Muscles of the uterus contract. The cervix opens (dilates). The waters break.

C The baby is pushed out through the vagina. The cord is tied and cut.

Stage 1 _____ Stage 2 _____ Stage 3 _____

What are “the waters”?

Workbook: fertility problems and solutions

Problems

Solutions

Low sperm count

Injecting sperm cell into egg cell, then IVF

Oviducts too narrow

Sperm ducts blocked

Lining stops implantation

Few ripe eggs released

Sperm do not swim well

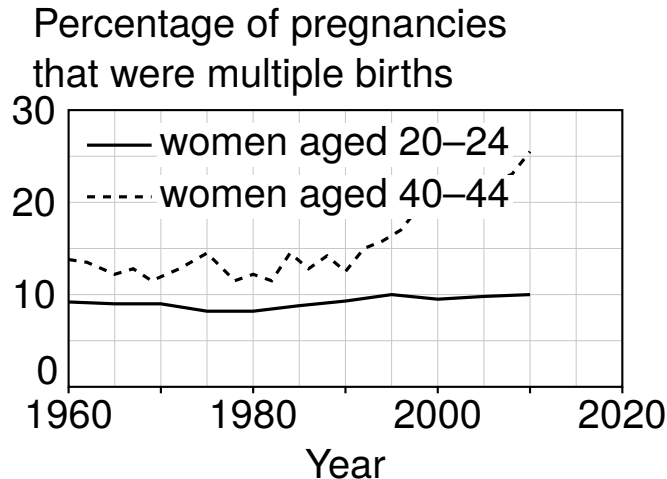
IVF: fertilisation in a dish

Put the cards in order.

- A** Eggs are collected from the woman's ovaries.
- B** An embryo is put into the uterus, where it may implant.
- C** The woman is given hormones, so several eggs mature.
- D** The eggs are mixed with sperm in a dish. Some are fertilised.
- E** The embryos grow for about two days.

Why are twins more common after IVF?

Workbook: question 3, describe



- (a) Describe what has happened to the number of multiple births in women aged 40-44.
- (c) Describe what happened to the number of multiple births in women aged 20-24.

Workbook: suggest and predict

Headlines: “Fertility treatment shown to increase chances of multiple birth” · “Older women more likely to have problems becoming pregnant” · “Fertility treatment becoming more common”

Q3 (b) Suggest a hypothesis for why this has been happening.

Q3 (d) Suggest a hypothesis for why this pattern is not like that for women aged 40–44.

Q4 (a) Predict the percentage of multiple births last year for women aged 40–44.

Spot the error (4)

Describe how a foetus gets oxygen and how it is protected in the uterus.

Sam's answer

The mother's blood flows through the umbilical cord into the foetus. Oxygen moves by diffusion from a low concentration to a high concentration. The amniotic fluid gives the foetus its food and protects it from bumps.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

- Q1** Name two substances that pass from the foetus's blood to the mother's blood. **[A]**
- Q2** Explain why a pregnant woman who smokes is likely to have a smaller baby. **[M]**
- Q3** IVF clinics now usually put one embryo into the uterus, not three. Suggest why, and explain how this changes the number of twins and triplets. **[E]**

Classwork

In-class work · Part D

- 1 Label the foetus in the uterus.
- 2 Explain how a waste product leaves the foetus.
- 3 Describe and explain a table about the foetus.

Name the substance, the direction and the process.

Ways to lose marks

Say the correct version of each one.

- 1 “A sperm cell has 46 chromosomes.”
- 2 “Fertilisation happens in the uterus.”
- 3 “Ovulation is when the lining is lost.”
- 4 “The mother’s blood flows into the foetus.”
- 5 “A sperm has a tail.” (asked to explain an adaptation)
- 6 “It went up because of IVF.” (asked to describe a graph)

Summary

- 1 Sex cells have 23 chromosomes; fertilisation in the oviduct makes 46.
- 2 Cycle: period, lining builds up, ovulation on day 14, lining kept thick.
- 3 Placenta: diffusion from high to low; the two bloods never mix.
- 4 Explain = fact + because. Describe a graph = trend + numbers.

Mini mock

In-class work · Part E

12 minutes of writing, no notes, no booklet.

Then we mark it together: A, M or E for each question.

Read the command word first: state, describe, explain, suggest.

Homework

Homework: two parts, about 40 minutes each

- 1** Part 1: sex cells, the two systems, the menstrual cycle.
- 2** Part 2: fertilisation, chromosomes, pregnancy, fertility.

Do the two parts on two different days.